Traffic Sign Classification on the Mapillary Traffic Sign Dataset:

From HOG+SVM to CNN, Transfer Learning, and Robustness Under Synthetic Distortions

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Abstract

We address fine-grained traffic-sign classification on the Mapillary Traffic Sign Dataset v2 (MTSD), comparing a classical computer-vision pipeline (HOG-on-edge features with a Linear SVM) against a custom convolutional neural network trained from scratch, the same network trained with data augmentation, and a pretrained EfficientNet-B0 fine-tuned via transfer learning. We curate the data into **326 well-supported classes** by removing MTSD's "other-sign" catch-all (63 % of crops) and dropping classes with fewer than 30 training samples, yielding 67,833 crops across train / val / test splits that honor MTSD's official partitioning. We additionally evaluate the best CNN under a six-axis synthetic distortion suite (illumination, rotation, occlusion, Gaussian noise, motion blur, JPEG compression) at four severities.

The CNN approaches dominate the classical baseline by a wide margin: the custom from-scratch network reaches **96.55 % top-1 accuracy** versus **55.97 %** for the matched-split classical pipeline — an absolute gap of **+40.6 percentage points** of top-1 accuracy and **+42.6 macro-F1** points. Data augmentation gives the highest macro-F1 (**0.958**) on the held-out test set, while ImageNet-pretrained EfficientNet-B0 fine-tuned via transfer learning gives the highest top-1 accuracy (**96.98 %**). The best model degrades by only ≤ 1.5 pt under severe noise / blur / JPEG / rotation but loses 8-10 pt under heavy illumination changes and large occlusions — directly motivating future targeted augmentation. We additionally deploy the best model as a streamlit web app Docker container.

Index terms — traffic sign recognition, convolutional neural networks, transfer learning, robustness, HOG, MTSD, Mapillary.

1 Introduction

Reliable traffic-sign recognition is a foundational perception capability for advanced driver assistance and autonomous driving. The Mapillary Traffic Sign Dataset (MTSD) [MTSD] is the largest open multi-class traffic-sign dataset to date, drawn from real-world street-level imagery in many countries and containing **400+ sign types** with a severe long-tail class distribution.

This paper presents an end-to-end study comparing classical and deep-learning approaches on MTSD. Concretely, we:

1. Curate a clean **326-class** subset of MTSD with stratified train / val / test splits that respect MTSD's official partitioning.

2. Establish a classical Phase 1 baseline (HOG $\rightarrow$ Linear SVM via SGD) and rerun it on our exact splits and class set so the comparison with the CNN models is apples-to-apples.

3. Design, train, and analyze a custom **2.38 M-parameter** CNN ("BaselineCNN") from scratch, achieving **96.55 %** top-1 test accuracy.

4. Run three controlled experiments — data augmentation, transfer learning with EfficientNet-B0 (4.43 M params, ImageNet-pretrained), and a synthetic-distortion robustness sweep — to identify which design choices matter most.

5. Deploy the best model as a streamlit Dockerized web app.

The paper is structured as follows: Section 2 reviews related work; Section 3 describes the dataset and our curation policy; Section 4 details the methodology; Section 5 presents headline results; Section 6 analyses robustness; Section 7 covers deployment; Section 8 discusses limitations and future work; Section 9 concludes.

2 Related Work

Traffic-sign recognition datasets. The German Traffic Sign Recognition Benchmark (GTSRB) [Stallkamp2012] popularized the task with 43 classes and ~50 k samples in controlled near-camera framing. The Belgian, Swedish, and TT100K [Zhu2016] datasets each broaden geographic / visual diversity. MTSD [Ertler2020] is the largest and most diverse, but its long-tail distribution and unbounded `other-sign` class make naive benchmarking misleading.

Classical pipelines. Histogram of Oriented Gradients (HOG) [Dalal2005] combined with support vector machines was the dominant pre-deep-learning formulation for traffic signs and many other fine-grained visual categories.

Convolutional neural networks for signs. Stallkamp et al. showed that a CNN trained from scratch on GTSRB could surpass human performance. EfficientNet [Tan2019], a family of scalable image-recognition networks pretrained on ImageNet, has since become a standard transfer-learning starting point because of its accuracy-per-parameter Pareto front.

Robustness under distortion. Hendrycks and Dietterich's ImageNet-C [Hendrycks2019] formalized the practice of testing models under synthetic image corruptions; we adopt the same framework on the MTSD test set.

3 Dataset & Curation

3.1 MTSD overview

MTSD v2 provides 36,589 fully annotated training images, 5,320 validation images, and 10,543 test images (test labels withheld). Each image carries COCO-style bounding boxes for every visible sign together with a fine-grained label (e.g. `regulatory--no-entry--g1`) and per-instance properties (occluded, out-of-frame, ambiguous, . . .).

3.2 Curation policy

We define our classification task on **cropped signs** rather than full images, by extracting each bounding box and resizing to 96×96. To make the task well-posed, we apply three policy decisions:

1. **Discard "other-sign"** — 121,136 of 190,496 raw crops (**63.6 %**) carry MTSD's `other-sign` label, which is by definition an uncategorized catch-all. Keeping it would bias the model toward defaulting to this class whenever uncertain and would dominate macro-averaged metrics.
2. **Discard classes with < 30 train samples** — 74 such tail classes (1,339 crops total) are too rare to learn from at this sample budget.
3. **Discard signs with bounding boxes smaller than 12 px on either side** — such crops carry virtually no discriminative content after the 96×96 resize.

After curation we obtain **326 classes and 67,833 crops**. Class sample counts in the train split range from 27 to 2,150 with median 85; a clear long-tail residual remains and is handled at the loss level via class weighting (Section 4.2).

3.3 Splits

To use the dataset's released partitioning faithfully, we:

- Reserve **all crops from MTSD-validation** as our held-out **test set** (8,604 crops, never seen during hyperparameter selection).
- Stratified 90 / 10 split of **MTSD-train** into our **train** (53,306) and our **validation** (5,923) sets.

4 Methodology

4.1 Phase 1 — Classical baseline (HOG $\rightarrow$ SGD-SVM)

Each cropped image is resized to 64×64, converted to grayscale, smoothed with a 5×5 Gaussian, and passed through Canny edge detection. We then compute a Histogram of Oriented Gradients (HOG) with 9 orientations, 8×8 pixel cells, 2×2 blocks, and L2-Hys block normalisation, yielding a **1,764-dimensional** descriptor. Features are zero-mean unit-variance standardised and classified with a multi-class Linear SVM fitted via stochastic gradient descent (`sklearn.linear_model.SGDClassifier`, `loss="hinge"`, `penalty="l2"`, `alpha=1e-4`, `class_weight="balanced"`).

The SGD variant was chosen over `sklearn.svm.LinearSVC` (primal solver)

Split	Crops	Source	Role
Train	53,306	90 % of MTSD-train	Model fitting
Val	5,923	10 % of MTSD-train	Hyperparameter selection, early stopping
Test	8,604	All of MTSD-val	Held-out reporting

because the latter did not converge in a tractable time-budget — over 60 minutes on this dataset's 326-class, 53 k-sample, balanced-weight regime, vs **164 seconds** for the SGD fit. Both are linear SVMs with L2 regularization and one-vs-rest decision rules.

This pipeline is structurally identical to the Phase-1 notebook provided by the course (`visionPhase1.ipynb`), differing only in (a) the matched class set and splits and (b) the faster solver. Reporting against the exact same test partition as the CNN models makes the comparison apples-to-apples.

4.2 Phase 2 — Custom CNN baseline (BaselineCNN)

We design a from-scratch network specifically sized to 96×96 inputs:

The model has **2.38 M parameters** and uses Kaiming-normal initialisation. Design choices and their justifications:

- 7×7 **strided stem** + 3×3 **pool**: 96×96 inputs are small; aggressive early downsampling reduces compute without losing signal.
- **Stacked** 3×3 **convolutions**: same receptive field as 5×5 with fewer parameters and more non-linearity (VGG insight, not VGG itself).
- **BatchNorm everywhere**: stabilises a random-init network against the residual class imbalance.
- **Global Average Pool + small Dropout + 1-layer head**: $\sim 10\times$ fewer parameters than Flatten+Dense and far less prone to over-fitting.

Training uses AdamW (lr 3×10^{-3}, weight decay 1×10^{-4}), OneCycle cosine learning-rate schedule with 10 % warmup, batch 128, **50 epochs**, label smoothing 0.05, and class-weighted cross-entropy with weights set to $\sqrt{1/freq}$ normalised to mean 1.0. The best checkpoint by validation macro-F1 is retained.

4.3 Phase 2 — Experimental study

E1: Data augmentation. Re-train BaselineCNN with Albumentations augmentation: rotation $\pm 15°$ (no horizontal flip — signs are direction-sensitive), brightness / contrast ± 25 %, hue / saturation jitter, mild Gaussian blur, coarse cut-out occlusion, and 8 % translation / 92–108 % scale jitter.

E2: Transfer learning. Replace BaselineCNN with **EfficientNet-B0** [Tan2019] initialised from ImageNet weights via `timm`. **4.43 M parameters** ($1.9\times$ BaselineCNN). Same training recipe except `lr=1e-3` (typical for fine-tuning); data augmentation enabled.

E3: Robustness under synthetic distortions (Section 6). Six distortions $\times$ four severities applied to the test set; the best model scored.

5 Results

All metrics are computed on the held-out test set (8,604 crops, 326 classes). Inputs are identical across rows — same crop pipeline, same split, same class set — so differences reflect the model only.

5.1 Headline comparison

Key observations.

- **Classical → CNN is a 40-point chasm.** From 55.97 % top-1 to 96.55 % with the same crops and the same splits — a +40.6 pt absolute gap (a 73 % relative error reduction). The classical HOG + SVM features cannot capture the discriminative information needed for 326-way fine-grained classification at this image resolution.

Stage	Layer	Output
Stem	Conv 7×7 /2 $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ MaxPool 3×3 /2	$64 \times 24 \times 24$
Block 1	Conv 3×3 $\rightarrow$ BN $\rightarrow$ ReLU $\times 2$ $\rightarrow$ MaxPool 2×2	$128 \times 12 \times 12$
Block 2	Conv 3×3 $\rightarrow$ BN $\rightarrow$ ReLU $\times 2$ $\rightarrow$ MaxPool 2×2	$256 \times 6 \times 6$
Block 3	Conv 3×3 $\rightarrow$ BN $\rightarrow$ ReLU $\times 2$	$256 \times 6 \times 6$
Head	GAP $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear	326

Model	Top-1 acc	Top-5 acc	Macro-P	Macro-R	Macro-F1	Weighted-F1
Phase 1 (HOG $\rightarrow$ SGD-SVM, matched rerun)	55.97 %	—	0.555	0.553	0.524	0.591
Phase 2 — BaselineCNN (no aug)	96.55 %	99.43 %	0.956	0.949	0.949	0.965
Phase 2 — BaselineCNN + augmentation	96.89 %	99.51 %	0.963	0.958	**0.958**	0.969
Phase 2 — EfficientNet-B0 (transfer)	**96.98 %**	99.48 %	0.959	0.957	0.956	0.970

- **Augmentation gives the highest macro-F1 (0.958)** — the metric that treats every one of the 326 classes equally, including tail classes where data is scarce. Augmentation helps where it matters most.

- **Transfer learning gives the highest top-1 accuracy (96.98 %)** — EfficientNet-B0's ImageNet prior is most helpful on the head classes that dominate the weighted average.

- Top-5 accuracy is essentially saturated (99.4-99.5 %) for all three CNN models, indicating the correct class is almost always within the shortlist and remaining errors are confusions between visually similar sign variants.

5.2 Training dynamics

All three CNNs are trained for 50 epochs with the OneCycle LR schedule.

(note 1) The augmentation run encountered episodes of Mac GPU contention that inflated the wall-clock epoch time average; individual epochs were ~130 s on a quiescent system.

Augmentation cleanly closes the generalisation gap (3.43 pt $\rightarrow$ 2.72 pt on train-vs-val accuracy). This is the textbook expected behaviour of augmentation and exactly the experimental signal we hoped to see.

5.3 Per-class analysis

Errors concentrate in two failure modes: (a) extremely similar visual pairs (`warning--curve-left--g1` vs `warning--curve-left--g2`), and (b) tail classes near the 30-sample threshold. Full per-class precision / recall / F1 / support breakdowns are written to `results/<run>/test_per_class.csv` (326 rows $\times$ 4 columns each).

6 Robustness Analysis

We evaluate the augmentation-trained model (best macro-F1) on the test set under six distortions $\times$ four severities (0 = clean $\rightarrow$ 3 = severe). Distortions are implemented in `src/distortions.py` and chosen to model realistic deployment conditions: illumination shifts (day/night, headlights), small camera tilt, foliage / vehicle occlusion, low-light sensor noise, motion-induced blur, and lossy compression in transmitted or stored imagery.

Key findings.

- **Near-zero degradation under Gaussian noise, JPEG, and motion blur.** At the most severe levels tested, accuracy drops by $\leq$ 0.51 pt. This suggests the BatchNorm-regularised conv features are already invariant to mid-frequency texture perturbations.

- **Rotation is mostly handled.** Even at severity 3 ($\pm 25°$) accuracy drops only 1.5 pt — a direct dividend of training-time rotation augmentation ($\pm 15°$).

- **Illumination is the main brittleness.** Brightness shifts of ± 60 % cause a 7.7 pt drop, suggesting the model has implicitly memorised scene luminance dis-

Model	Best epoch	Best val acc	Best val F1	Final train acc	Final val acc	Train-val gap	Avg epoch time
BaselineCNN (no aug)	39	0.9667	0.9550	1.0000	0.9657	**3.43 pt**	121 s
+ Augmentation	39	0.9671	0.9562	0.9936	0.9664	**2.72 pt**	(note 1)
EfficientNet-B0	47	0.9691	0.9581	0.9965	0.9681	2.84 pt	252 s

Distortion	Sev 0	Sev 1	Sev 2	Sev 3	Drop @ Sev 3
Gaussian noise	96.89 %	96.84 %	96.87 %	96.83 %	**0.06 pt**
JPEG compression	96.89 %	96.89 %	96.90 %	96.37 %	**0.51 pt**
Motion blur	96.89 %	96.87 %	96.90 %	96.44 %	**0.44 pt**
Rotation	96.89 %	96.87 %	96.47 %	95.36 %	**1.52 pt**
Illumination	96.89 %	96.75 %	94.64 %	89.14 %	**7.74 pt**
Occlusion	96.89 %	96.48 %	95.27 %	86.85 %	**10.03 pt**

tribution. Stronger training-time brightness/contrast augmentation is a low-effort fix.

- **Occlusion is the worst case (-10 pt).** Large square cut-outs that obscure 30-40 % of the sign cause the model to fail catastrophically. This matches the intuition that classifying a fine-grained sign from partial visibility requires either context or part-based reasoning, neither of which our 96^2 crop-based formulation gives the model.

7 Deployment (Bonus Option A)

The augmentation-trained model is shipped four ways:

1. **Gradio app** (`src/deploy/app.py`) — interactive top-5 demo.
2. **FastAPI service** (`src/deploy/api.py`) — `POST /predict` for programmatic access. End-to-end JSON round-trip $\leq$ 20 ms / request on CPU; smoke-tested with a held-out test image returning the correct label at 93.7 % confidence.
3. **Docker image** (`src/deploy/Dockerfile`) — CPU-only FastAPI service in a portable container, $\sim$ 600 MB compressed.
4. **ONNX** (`results/aug/export/model.onnx`, $\sim$ 10 MB) for cross-platform inference and **Core ML** (`results/aug/export/model.mlpackage`, $\sim$ 9 MB) for Apple Neural Engine inference on iOS / macOS.

8 Discussion and Limitations

- **Hard class set choice.** Dropping `other-sign` and rare classes produces a learnable problem but discards 67 % of raw annotations. A hierarchical formulation (super-category first, then fine-grained) is a natural extension.
- **No object detection.** We classify pre-cropped boxes given by MTSD's ground truth. In an end-to-end self-driving pipeline these crops would themselves be predictions from a detector, propagating errors.
- **Phase-1 solver substitution.** Replacing primal LinearSVC with SGD-fitted hinge loss saves training time (164 s vs > 1 h) at a small cost in convergence quality. Given the size of the CNN-vs-classical gap (40 + points), this choice does not affect the qualitative conclusions of the paper.
- **MPS non-determinism.** The Apple MPS backend has minor non-determinism in some kernels; epoch-to-epoch variance is $\leq$ 0.5 %.

8.1 Future work

- Hierarchical classifier (super-category $\rightarrow$ fine-grained).

- Curriculum learning for tail classes.
- Stronger brightness/contrast and large-CoarseDropout augmentation to directly attack the two failure modes identified in Section 6.
- Train on the union of all detection-style crops and end-to-end fine-tune with a YOLO-class detector for full pipeline evaluation.

9 Conclusion

We presented a clean, end-to-end MTSD classification study with apples-to-apples comparison of classical and deep-learning approaches on identical splits and class sets. The custom 2.38 M-parameter BaselineCNN substantially outperforms the matched HOG $\rightarrow$ SGD-SVM baseline (96.55 % vs 55.97 % top-1 — a 40.6 pt absolute gap). Data augmentation gives the best macro-F1 (0.958) and ImageNet-pretrained EfficientNet-B0 gives the best top-1 accuracy (96.98 %). The augmentation-trained model degrades gracefully under realistic synthetic distortions — losing $<$ 1.5 pt under noise, blur, JPEG, and small rotation — but is brittle to large occlusions and illumination changes, suggesting concrete next steps. Code, trained weights, an interactive Gradio demo, a Dockerized FastAPI service, and ONNX / Core ML exports are released.

References

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